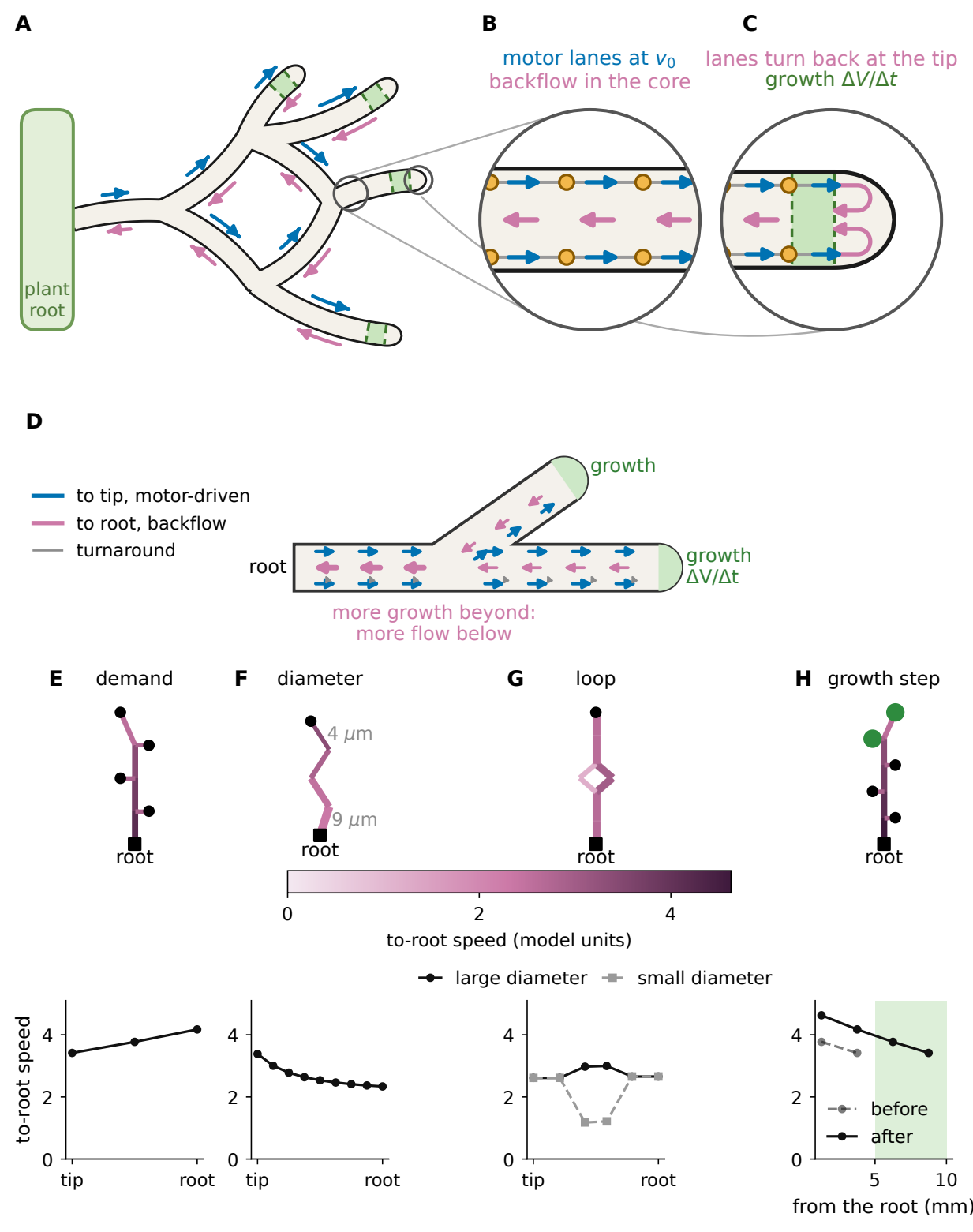

A
B
C
motor lanes at v_0
backflow in the core
lanes turn back at the tip
growth $\Delta V/\Delta t$
plant
root
D
to tip, motor-driven
to root, backflow
turnaround
growth
root
growth
$\Delta V/\Delta t$
more growth beyond:
more flow below
E
demand
root
F
diameter
4 μm
9 μm
root
G
loop
root
H
growth step
root
0
2
4
to-root speed (model units)
large diameter
small diameter
to-root speed
0
2
4
tip
root
tip
root
tip
root
before
after
5
10
from the root (mm)